Lab 15 – Static Malware Analysis Report
CS 362 – Digital Forensics
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This lab focused on performing static malware analysis on several Windows executable samples. The main goal was to understand how malware is structured, which Windows API functions it relies on, what resources it contains, and what information can be extracted without actually executing the program. By using tools like pefile, strings, wrestool, and online analysis platforms such as VirusTotal and Hybrid Analysis, I was able to identify characteristics that reveal how the malware behaves.

PE Header Analysis

Using the Python pefile library, I examined ircbot.exe to view its DOS header, NT header, and optional header. The DOS header confirmed the standard “MZ” signature, and the NT header validated the file as a Portable Executable with the correct PE signature. The optional header showed details such as the 32-bit architecture, entry point address, image base, and memory layout. These header values help describe how the executable loads into memory and where its code and data live.

Section Analysis

The malware sample contained the typical set of PE sections, including .text, .rdata, .data, .idata, and .reloc. Each section serves a different purpose inside the executable. For example, .text holds the program’s code, .rdata stores constants, and .idata is where all imported Windows API functions are listed. Examining these sections helps determine what the malware may attempt to do and how complex it is.

Import Table Analysis

Looking at the import table revealed many Windows API functions that point to network communication, system modification, and user interaction. Some examples included HttpSendRequestA, InternetGetConnectedState, CreateFileA, WriteFile, and GetUserNameA. The presence of these functions suggests the malware may send data over the internet, manipulate files, interact with processes, or collect information from the system. These are typical behaviors seen in botnet malware and backdoors.

Resource Extraction

Using wrestool and icotool, I attempted to extract icons and other embedded resources from the samples. While ircbot.exe did not contain any resources, fakepdfmalware.exe included a disguised PDF-style icon. This kind of deception is common in malware designed to trick users

into opening a file that appears to be a document. Identifying embedded icons helps reveal how malware presents itself to victims.

Strings Analysis

Analyzing printable strings inside ircbot.exe provided clear signs of malicious intent. I found references to HTTP communication, server commands, passwords, login names, and IRC-related strings such as bot names and channel information. The presence of words like “server,” “bot,” “pass,” “recv,” and “send” points toward command-and-control communication. These strings also showed that the malware attempts to connect to remote servers, send and receive data, and possibly join IRC channels used for botnet control.

Malware Intelligence Results

I submitted the malware sample to both VirusTotal and Hybrid Analysis.
VirusTotal showed that a large majority of antivirus engines detected the file as malicious, with labels such as Trojan, Rbot, Sdbot, and “Backdoor.” This confirms the sample is part of a known botnet family.
Hybrid Analysis labeled it as malicious with a high threat score and displayed behavioral indicators such as network communication, file system activity, and suspicious API calls. These results match what was observed during static analysis.

This lab demonstrated how much information can be obtained from malware without running it. By analyzing headers, sections, imports, resources, strings, and threat intelligence reports, I was able to clearly identify typical characteristics of botnet malware. Static analysis is an essential skill in digital forensics because it helps understand how malware works internally and allows investigators to gather evidence safely. The samples used in this lab showed clear signs of command-and-control behavior, disguised icons, and malicious network activity.

VirusTotal - File - 15c397 | Free Automated Malwar

https://hybrid-analysis.com/sample/15c3973bc94a4e61e0e20c38250394fbd799182e833ef491954f7a71a2eb7280

OffSec | Kali Linux | Kali Tools | Kali Docs | Kali Forums | Kali NetHunter | Exploit-DB | Google Hacking DB

HYBRID ANALYSIS | Sandbox | Quick Scans | File Collections | Resources | Request Info | IP, Domain, Hash...

Analysis Overview

Request Report Deletion | Show Sample Content

Submission name:	ircbot.exe
Size:	243KiB
Type:	peexe executable
Mime:	application/x-dosexec
SHA256:	15c3973bc94a4e61e0e20c38250394fbd799182e833ef491954f7a71a2eb7280
Submitted At:	2018-09-09 09:36:17 (UTC)
Last Anti-Virus Scan:	2025-04-08 22:11:46 (UTC)
Last Sandbox Report:	2025-04-08 22:11:45 (UTC)

malicious

Threat Score: 100/100
AV Detection: 83%
Labeled As: Sdbot.Generic

Post | Link | E-Mail

0 Community Score 0

Analysis Overview
Anti-Virus Scanner Results
Falcon Sandbox Reports (7)
Relations
Incident Response
Community (1)

Back to top

Anti-Virus Results

Updated 7 months ago - Missing Results - Click to Refresh

CrowdStrike Falcon
Static Analysis and ML
Malicious (100%)
No Additional Data

MetaDefender
Multi Scan Analysis
Malicious (16/24)
More Details

Be alerted when interesting malware samples emerge. Every day, MalQuery ingests over 750,000 malware samples, you can use YARA to monitor MalQuery as new samples are added and be alerted via email or API when there is a match.

Learn more

VirusTotal - File - 15c397

https://www.virustotal.com/gui/file/15c3973bc94a4e61e0e20c38250394fbd799182e833ef491954f7a71a2eb7280

OffSec | Kali Linux | Kali Tools | Kali Docs | Kali Forums | Kali NetHunter | Exploit-DB | Google Hacking DB

15c3973bc94a4e61e0e20c38250394fbd799182e833ef491954f7a71a2eb7280 | Sign in

51 / 71
Community Score

51/71 security vendors flagged this file as malicious

Reanalyze | Similar | More

15c3973bc94a4e61e0e20c38250394fbd799182e833ef491954f7a71a2eb7280
ircbot.exe
peexe | checks-user-input | detect-debug-environment | spreader

Size 242.50 KB | Last Analysis Date 8 months ago | EXE

DETECTION | DETAILS | RELATIONS | BEHAVIOR | COMMUNITY 8

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label: trojan.rbot/sdbot | Threat categories: trojan | Family labels: rbot sdbot ircbot

Security vendors' analysis — Do you want to automate checks?

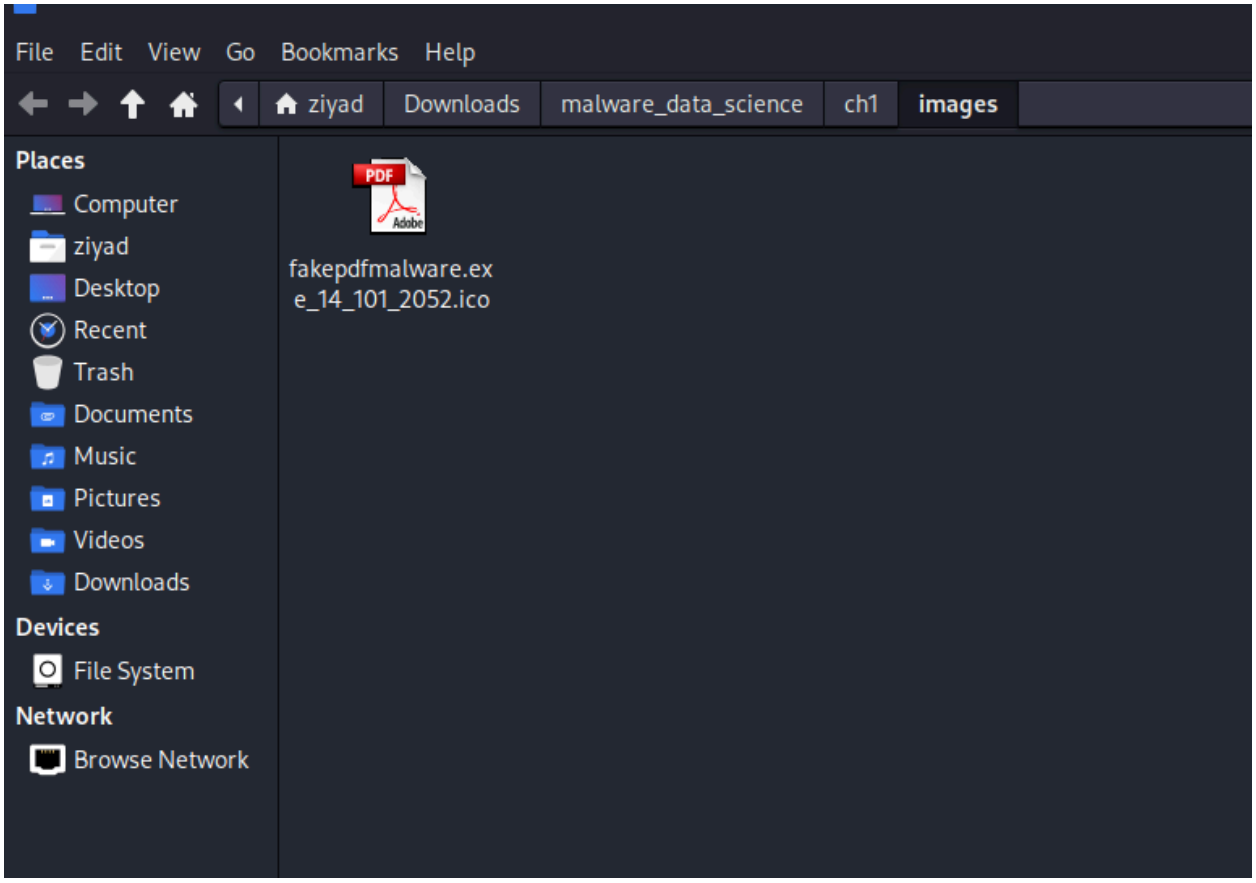
AhnLab-V3	Trojan/Win32.SdBot.C2880084	Alibaba	Backdoor:Win32/IRCBot.d11aadb7
AliCloud	Backdoor:Win/IRCBot.BZ	Antiy-AVL	Trojan[Backdoor]/Win32.Rbot
Arcabit	Generic.Sdbot.3C3B4586	Arctic Wolf	Unsafe
Avast	Win32:Rbot-DTQ [Trj]	AVG	Win32:Rbot-DTQ [Trj]
BitDefender	Generic.Sdbot.3C3B4586	Bkav Pro	W32.AIDetectMalware
ClamAV	Win.Trojan.Mybot-4324	CrowdStrike Falcon	Win/malicious_confidence_100% (W)
CTX	Exe.trojan.generic	Cynet	Malicious (score: 100)

File Actions Edit View Help

```
HttpSendRequestA
HttpOpenRequestA
InternetGetConnectedStateEx
InternetGetConnectedState
recvfrom
recv
sendto
send
connect
GetUserNameA
SendMessageA
user32.dll
User32.dll failed. <%d>
winpass
sqlpassoainstall
databasepassword
databasepass
dbpassword
dbpass
domainpassword
domainpass
loginpass
oemuser
user
homeuser
server
pass
pass1234
passwd
password
password1
[MAIN]: Connected to %s.
USER %s 0 0 :%s
PASS %s
httpcon
[HTTPD]: Failed to start server thread, error: <%d>.
[HTTPD]: Server listening on IP: %s:%d, Directory: %s\.
http
httpserver
[MAIN]: User %s logged out.
USERHOST %s
KBOT5
_CrtIsValidHeapPointer(pUserData)
szUserMessage ≠ NULL
bbot
bBot-Version 0.6
botirc.net
irc.server2.net
channelpass2
bBot|
USER32.dll
```

File Actions Edit View Help

```
`.rdata
@.data
.idata
.reloc
DSVW
h,@C
TSVW
Ph@@C
(_^[
Y_^[
PSVW
hh@C
DSVW
ht@C
ht@C
Y_^[
DSVW
HSVW
X_^[
Pj,h
Pj,h
Pj`hA
Pj`h@
PSVW
PSVW
Ph\GC
Ph(GC
$hxFC
PhpFC
PhTFC
DSVW
@SVW
DSVW
XSVW
YYPh
YYPh
YYPh
YPhTHC
Y_^[
LSVW
Y_^[
Ph~f
Ph~f
hPKC
h0KC
Ph\KC
htPC
hpOC
h|NC
PhXNC
Ph<NC
Ph\MC
Ph@MC
:
```

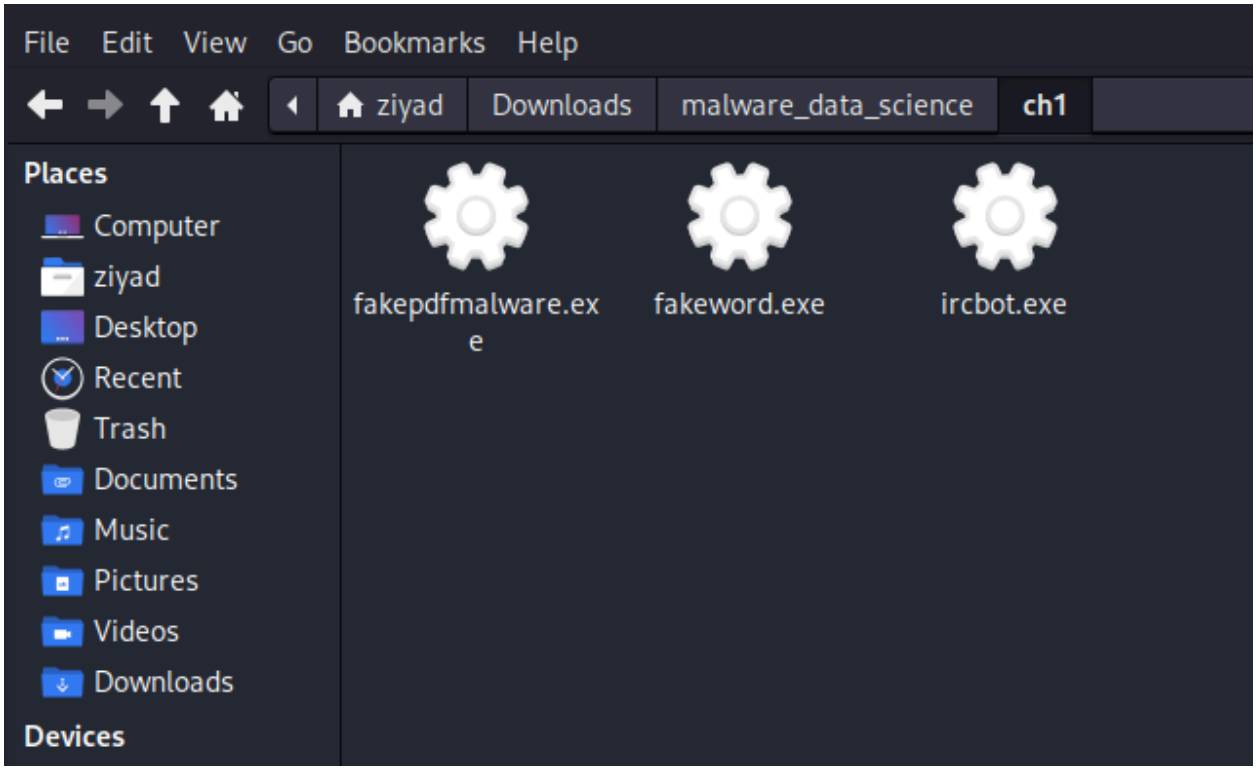



```
┌──(ziyad㉿kali)-[~/Downloads/malware_data_science/ch1]
└─$ wrestool --raw -o images fakepdfmalware.exe
--type='FILE' --name=102 --language=2052 [offset=0×8dc0 size=276994]
--type=3 --name=1 --language=2052 [type=icon offset=0×7100 size=7336]
--type=14 --name=101 --language=2052 [type=group_icon offset=0×8da8 size=20]

┌──(ziyad㉿kali)-[~/Downloads/malware_data_science/ch1]
└─$ wrestool --raw -o images fakeword.exe
--type='CUSTOM' --name=101 --language=1033 [offset=0×14c724 size=53248]
--type='CUSTOM' --name=102 --language=1033 [offset=0×103924 size=298496]
--type='CUSTOM' --name=103 --language=1033 [offset=0×e98f0 size=106548]
--type=3 --name=30001 --language=0 [type=icon offset=0×e9608 size=744]
--type=3 --name=30002 --language=0 [type=icon offset=0×e94e0 size=296]
--type=3 --name=30003 --language=0 [type=icon offset=0×e8638 size=3752]
--type=14 --name=1 --language=0 [type=group_icon offset=0×e8608 size=48]
--type=16 --name=1 --language=3082 [type=version offset=0×e8200 size=1032]

┌──(ziyad㉿kali)-[~/Downloads/malware_data_science/ch1]
└─$ wrestool --raw -o images ircbot.exe
wrestool: ircbot.exe: file contains no resources

┌──(ziyad㉿kali)-[~/Downloads/malware_data_science/ch1]
└─$
```

```
┌──(ziyad㉿kali)-[~/Downloads/malware_data_science/ch1]
└─$ sudo apt update
Get:1 http://us.mirror.ionos.com/linux/distributions/kali/kali kali-rolling InRelease [34.0 kB]
Get:2 http://us.mirror.ionos.com/linux/distributions/kali/kali kali-rolling/main amd64 Packages [21.0 MB]
Get:3 http://us.mirror.ionos.com/linux/distributions/kali/kali kali-rolling/main amd64 Contents (deb) [52.7 MB]
Get:4 http://us.mirror.ionos.com/linux/distributions/kali/kali kali-rolling/contrib amd64 Packages [115 kB]
Get:5 http://us.mirror.ionos.com/linux/distributions/kali/kali kali-rolling/contrib amd64 Contents (deb) [260 kB]
Get:6 http://us.mirror.ionos.com/linux/distributions/kali/kali kali-rolling/non-free amd64 Packages [187 kB]
Get:7 http://us.mirror.ionos.com/linux/distributions/kali/kali kali-rolling/non-free amd64 Contents (deb) [894 kB]
Get:8 http://us.mirror.ionos.com/linux/distributions/kali/kali kali-rolling/non-free-firmware amd64 Packages [11.7 kB]
Get:9 http://us.mirror.ionos.com/linux/distributions/kali/kali kali-rolling/non-free-firmware amd64 Contents (deb) [28.5 kB]
Fetched 75.2 MB in 8s (9,066 kB/s)
1537 packages can be upgraded. Run 'apt list --upgradable' to see them.

┌──(ziyad㉿kali)-[~/Downloads/malware_data_science/ch1]
└─$ sudo apt install icoutils
Installing:
  icoutils

Suggested packages:
  libterm-readline-gnu-perl  | libterm-readline-perl-perl

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1537
  Download size: 65.4 kB
  Space needed: 226 kB / 28.7 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 icoutils amd64 0.32.3-6 [65.4 kB]
Fetched 65.4 kB in 0s (234 kB/s)
Selecting previously unselected package icoutils.
(Reading database ... 411917 files and directories currently installed.)
Preparing to unpack .../icoutils_0.32.3-6_amd64.deb ...
Unpacking icoutils (0.32.3-6) ...
Setting up icoutils (0.32.3-6) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.2.7) ...

┌──(ziyad㉿kali)-[~/Downloads/malware_data_science/ch1]
└─$
```

```
ziyad@kali: ~/Downloads/malware_data_science/ch1
File  Actions  Edit  View  Help
    \n0x39C00    0x0   Characteristics:              0x0       \n0x39C04
0x4   TimeDateStamp:                0x0        [Thu Jan  1 00:00:00 1970 UTC
]\n0x39C08    0x8   ForwarderChain:              0x0       \n0x39C0C    0xC
   Name:                          0x96424   \n0x39C10    0x10  FirstThunk:
                 0x96230   \n\nKERNEL32.DLL.GetLocalTime Hint[0]\nKERNEL32.DL
L.ExitThread Hint[0]\nKERNEL32.DLL.CloseHandle Hint[0]\nKERNEL32.DLL.WriteFil
e Hint[0]\nKERNEL32.DLL.CreateFileA Hint[0]\nKERNEL32.DLL.ExitProcess Hint[0]
\nKERNEL32.DLL.CreateProcessA Hint[0]\nKERNEL32.DLL.GetTickCount Hint[0]\nKER
NEL32.DLL.GetModuleFileNameA Hint[0]\nKERNEL32.DLL.GetSystemDirectoryA Hint[0
]\nKERNEL32.DLL.Sleep Hint[0]\nKERNEL32.DLL.GetTimeFormatA Hint[0]\nKERNEL32.
DLL.GetDateFormatA Hint[0]\nKERNEL32.DLL.GetLastError Hint[0]\nKERNEL32.DLL.C
reateThread Hint[0]\nKERNEL32.DLL.GetFileSize Hint[0]\nKERNEL32.DLL.GetFileAt
tributesA Hint[0]\nKERNEL32.DLL.FindClose Hint[0]\nKERNEL32.DLL.FileTimeToSys
temTime Hint[0]\nKERNEL32.DLL.FileTimeToLocalFileTime Hint[0]\nKERNEL32.DLL.F
indNextFileA Hint[0]\nKERNEL32.DLL.FindFirstFileA Hint[0]\nKERNEL32.DLL.ReadF
ile Hint[0]\nKERNEL32.DLL.SetFilePointer Hint[0]\nKERNEL32.DLL.WriteConsoleA
Hint[0]\nKERNEL32.DLL.GetStdHandle Hint[0]\nKERNEL32.DLL.LoadLibraryA Hint[0]
\nKERNEL32.DLL.GetProcAddress Hint[0]\nKERNEL32.DLL.GetModuleHandleA Hint[0]\
nKERNEL32.DLL.FormatMessageA Hint[0]\nKERNEL32.DLL.GlobalUnlock Hint[0]\nKERN
EL32.DLL.GlobalLock Hint[0]\nKERNEL32.DLL.UnmapViewOfFile Hint[0]\nKERNEL32.D
LL.MapViewOfFile Hint[0]\nKERNEL32.DLL.CreateFileMappingA Hint[0]\nKERNEL32.D
LL.SetFileTime Hint[0]\nKERNEL32.DLL.GetFileTime Hint[0]\nKERNEL32.DLL.Expand
EnvironmentStringsA Hint[0]\nKERNEL32.DLL.SetFileAttributesA Hint[0]\nKERNEL3
2.DLL.GetTempPathA Hint[0]\nKERNEL32.DLL.GetCurrentProcess Hint[0]\nKERNEL32.
DLL.TerminateProcess Hint[0]\nKERNEL32.DLL.OpenProcess Hint[0]\nKERNEL32.DLL.
GetComputerNameA Hint[0]\nKERNEL32.DLL.GetLocaleInfoA Hint[0]\nKERNEL32.DLL.G
etVersionExA Hint[0]\nKERNEL32.DLL.TerminateThread Hint[0]\nKERNEL32.DLL.Flus
hFileBuffers Hint[0]\nKERNEL32.DLL.SetStdHandle Hint[0]\nKERNEL32.DLL.IsBadWr
itePtr Hint[0]\nKERNEL32.DLL.IsBadReadPtr Hint[0]\nKERNEL32.DLL.HeapValidate
Hint[0]\nKERNEL32.DLL.GetStartupInfoA Hint[0]\nKERNEL32.DLL.GetCommandLineA H
int[0]\nKERNEL32.DLL.GetVersion Hint[0]\nKERNEL32.DLL.DebugBreak Hint[0]\nKER
NEL32.DLL.InterlockedDecrement Hint[0]\nKERNEL32.DLL.OutputDebugStringA Hint[
0]\nKERNEL32.DLL.InterlockedIncrement Hint[0]\nKERNEL32.DLL.HeapAlloc Hint[0]
\nKERNEL32.DLL.HeapReAlloc Hint[0]\nKERNEL32.DLL.HeapFree Hint[0]\nKERNEL32.D
LL.HeapDestroy Hint[0]\nKERNEL32.DLL.HeapCreate Hint[0]\nKERNEL32.DLL.Virtual
Free Hint[0]\nKERNEL32.DLL.VirtualAlloc Hint[0]\nKERNEL32.DLL.WideCharToMulti
Byte Hint[0]\nKERNEL32.DLL.MultiByteToWideChar Hint[0]\nKERNEL32.DLL.LCMapStr
ingA Hint[0]\nKERNEL32.DLL.LCMapStringW Hint[0]\nKERNEL32.DLL.GetCPInfo Hint[
0]\nKERNEL32.DLL.GetACP Hint[0]\nKERNEL32.DLL.GetOEMCP Hint[0]\nKERNEL32.DLL.
UnhandledExceptionFilter Hint[0]\nKERNEL32.DLL.FreeEnvironmentStringsA Hint[0
]\nKERNEL32.DLL.FreeEnvironmentStringsW Hint[0]\nKERNEL32.DLL.GetEnvironmentS
trings Hint[0]\nKERNEL32.DLL.GetEnvironmentStringsW Hint[0]\nKERNEL32.DLL.Set
HandleCount Hint[0]\nKERNEL32.DLL.GetFileType Hint[0]\nKERNEL32.DLL.RtlUnwind
 Hint[0]\nKERNEL32.DLL.SetConsoleCtrlHandler Hint[0]\nKERNEL32.DLL.GetStringT
ypeA Hint[0]\nKERNEL32.DLL.GetStringTypeW Hint[0]\nKERNEL32.DLL.SetEndOfFile
Hint[0]\n\n[IMAGE_IMPORT_DESCRIPTOR]\n0x39C14    0x0   OriginalFirstThunk:
        0x0          \n0x39C14    0x0   Characteristics:              0x0
  \n0x39C18    0x4   TimeDateStamp:                0x0        [Thu Jan  1 00
:00:00 1970 UTC]\n0x39C1C    0x8   ForwarderChain:              0x0       \
n0x39C20    0xC   Name:                          0x96431   \n0x39C24    0x10
 FirstThunk:                    0x963F4   \n\nUSER32.dll.MessageBoxA Hint[0]\
n\n
>>>
```

```
File  Actions  Edit  View  Help
┌──(ziyad㉿kali)-[~/Downloads/malware_data_science/ch1]
└─$ python3
Python 3.13.3 (main, Apr 10 2025, 21:38:51) [GCC 14.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import pefile
>>> pe = pefile.PE("ircbot.exe")
>>> pe.dump_info()
"----------Parsing Warnings----------\n\nAddressOfEntryPoint lies outside the
 sections' boundaries. AddressOfEntryPoint: 0xcc00ffee\n\n----------DOS_HEADE
R----------\n\n[IMAGE_DOS_HEADER]\n0x0        0x0   e_magic:
      0x5A4D    \n0x2        0x2   e_cblp:                        0x90      \
n0x4      0x4   e_cp:                          0x3       \n0x6        0x6
 e_crlc:                          0x0       \n0x8        0x8   e_cparhdr:
              0x4       \n0x A        0xA   e_minalloc:                    0x
0      \n0xC        0xC   e_maxalloc:                    0xFFFF    \n0xE
    0xE   e_ss:                          0x0       \n0x10       0x10  e_sp:
                 0xB8      \n0x12       0x12  e_csum:
      0x0       \n0x14       0x14  e_ip:                          0x0
\n0x16       0x16  e_cs:                          0x0       \n0x18       0x18
  e_lfarlc:                       0x40      \n0x1A       0x1A  e_ovno:
                0x0       \n0x1C       0x1C  e_res:                        \
n0x24       0x24  e_oemid:                       0x0       \n0x26       0x26
 e_oeminfo:                       0x0       \n0x28       0x28  e_res2:
                \n0x3C       0x3C  e_lfanew:                      0xE0      \n
\n----------NT_HEADERS----------\n\n[IMAGE_NT_HEADERS]\n0xE0       0x0   Sign
ature:                     0x4550    \n\n----------FILE_HEADER----------\n\n[
IMAGE_FILE_HEADER]\n0xE4       0x0   Machine:                      0x14C
 \n0xE6       0x2   NumberOfSections:              0x5       \n0xE8       0x4
   TimeDateStamp:                 0x4F79D506 [Mon Apr  2 16:34:14 2012 UTC]\n
0xEC       0x8   PointerToSymbolTable:          0x0       \n0xF0       0x C
NumberOfSymbols:               0x0       \n0xF4       0x10  SizeOfOptionalHea
der:          0xE0      \n0xF6       0x12  Characteristics:                0x1
0F     \nFlags: IMAGE_FILE_32BIT_MACHINE, IMAGE_FILE_EXECUTABLE_IMAGE, IMAGE_
FILE_LINE_NUMS_STRIPPED, IMAGE_FILE_LOCAL_SYMS_STRIPPED, IMAGE_FILE_RELOCS_ST
RIPPED\n\n----------OPTIONAL_HEADER----------\n\n[IMAGE_OPTIONAL_HEADER]\n0xF
8       0x0   Magic:                         0x10B     \n0xFA       0x2   Maj
orLinkerVersion:            0x6       \n0xFB       0x3   MinorLinkerVersion:
           0x0       \n0xFC       0x4   SizeOfCode:                     0x32A0
0   \n0x100      0x8   SizeOfInitializedData:         0x64200   \n0x104
0xC   SizeOfUninitializedData:       0x0       \n0x108      0x10  AddressOfEn
tryPoint:           0xCC00FFEE\n0x10C      0x14  BaseOfCode:
   0x1000    \n0x110      0x18  BaseOfData:                    0x1000    \n0x
114      0x1C  ImageBase:                     0x400000  \n0x118      0x20  Se
ctionAlignment:              0x1000    \n0x11C      0x24  FileAlignment:
            0x200     \n0x120      0x28  MajorOperatingSystemVersion:   0x4
     \n0x122      0x2A  MinorOperatingSystemVersion:   0x0       \n0x124
 0x2C  MajorImageVersion:             0x0       \n0x126      0x2E  MinorImage
Version:             0x0       \n0x128      0x30  MajorSubsystemVersion:
    0x4       \n0x12A      0x32  MinorSubsystemVersion:         0x0       \n0
x12C      0x34  Reserved1:                     0x0       \n0x130      0x38  S
izeOfImage:                  0x9A000   \n0x134      0x3C  SizeOfHeaders:
             0x1000    \n0x138      0x40  CheckSum:                        0x0
      \n0x13C      0x44  Subsystem:                     0x2       \n0x13E
  0x46  DllCharacteristics:            0x0       \n0x140      0x48  SizeOfSta
```


File
Edit
View
Go
Bookmarks
Help
ziyad
Downloads
malware_data_science
ch1
Places
Computer
ziyad
Desktop
Recent
Trash
Documents
Music
Pictures
Videos
Downloads
Devices
fakepdfmalware.exe
fakeword.exe
ircbot.exe